THE AUTOCORRELATION FORMULA

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Abstract. We derive an exact character-sum formula for the cyclic autocorrelation $R(\ell)$ of the base repetend of $1/p$. For primes where b is a primitive root, the formula is

$$R(\ell) = \frac{1}{p}\sum_{k=0}^{p-1} G(k,\, -kb^{-\ell} \bmod p),$$

where $G(k,k') = \sum_d \widehat{\mathbf{1}}_{B_d}(k)\,\widehat{\mathbf{1}}_{B_d}(k')$ is the cross-spectral function of the bin partition. For primitive-root primes, this closes the formula chain from the digit function to the eigenvalue spectrum: the bin sizes determine G, G determines R, and R determines the eigenvalues via the DFT.

The derivation uses the orthogonality of additive characters on $\mathbb{Z}/p\mathbb{Z}$ [5, 7] and the identity $\sum_{k,k'} G(k,k') = 0$, which forces the off-diagonal correction to cancel exactly. The cross-spectral function G extends the spectral power $\Phi(k) = G(k,-k)$ from [4] to a two-variable object encoding both magnitude and phase information of the bin Fourier transforms.

1. Introduction

The cross-alignment matrix $\mathbf{A}(p)$ of a prime p in base b encodes the digit-equality structure of the fractional field [2]. Its eigenvalue spectrum, which records the harmonic content of the field, was identified in [3]: for primes where b is a primitive root modulo p, the eigenvalues are the discrete Fourier transform of the cyclic autocorrelation $R(\ell)$ of the base repetend $1/p$. The autocorrelation counts position-wise digit matches between the repetend and its cyclic rotations.

The spectral power paper [4] computed $\Phi(k) = \sum_d |\widehat{\mathbf{1}}_{B_d}(k)|^2$ in closed form as a sum of squared Dirichlet kernels weighted by the bin sizes. This gave a complete description of the magnitude data of the bin Fourier coefficients but did not determine the autocorrelation, since $R(\ell)$ depends on the phase information that the squaring operation discards. A gap remained: expressing $R(\ell)$ itself through the Fourier data of the bin partition.

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This paper closes the gap. We introduce the *cross-spectral function* $G(k,k') = \sum_d \widehat{\mathbf{1}}_{B_d}(k)\,\widehat{\mathbf{1}}_{B_d}(k')$ of the bin partition, a two-variable extension of Φ whose antidiagonal $G(k,-k)$ recovers $\Phi(k)$ and whose off-diagonal entries carry the phase information needed for $R(\ell)$. We prove (Theorem 3) that for primes where b is a primitive root,

$$R(\ell) = \frac{1}{p}\sum_{k=0}^{p-1} G\big(k,\, -k\, b^{-\ell} \bmod p\big)\,,$$

expressing the autocorrelation as a character sum over G along an orbit path determined by the shift ℓ. The proof rests on a single algebraic identity: $\sum_{k,k'} G(k,k') = 0$ (Lemma 2), which forces a cancellation in the naive expansion that collapses two terms into one.

Combined with [1, 3, 4], the autocorrelation formula closes the chain from the bin sizes of the digit function to the eigenvalue spectrum of the cross-alignment matrix in finitely many explicit steps. Every step is computable from p and b alone, with no empirical counting required. The multi-orbit case (where b is not a primitive root) is not handled by the present technique and remains open; we discuss the obstruction in §6.

2. The Cross-Spectral Function

Definition 1. The *cross-spectral function* of the bin partition is

$$G(k,k') = \sum_{d=0}^{b-1} \widehat{\mathbf{1}}_{B_d}(k)\,\widehat{\mathbf{1}}_{B_d}(k'),$$

where $\widehat{\mathbf{1}}_{B_d}(k) = \sum_{r\in B_d} e^{-2\pi i k r/p}$ is the additive Fourier transform of the d-th bin indicator.

The antidiagonal $G(k,-k) = \sum_d |\widehat{\mathbf{1}}_{B_d}(k)|^2 = \Phi(k)$ recovers the spectral power from [4] (using $\widehat{\mathbf{1}}_{B_d}(-k) = \overline{\widehat{\mathbf{1}}_{B_d}(k)}$ for real-valued indicators). The off-diagonal entries encode phase relationships between bins at different frequencies.

Lemma 2. $\sum_{k=0}^{p-1}\sum_{k'=0}^{p-1} G(k,k') = 0$.

Proof. By definition,

$$\sum_{k,k'} G(k,k') = \sum_{k,k'}\sum_d \widehat{\mathbf{1}}_{B_d}(k)\,\widehat{\mathbf{1}}_{B_d}(k') = \sum_d \left(\sum_{k=0}^{p-1} \widehat{\mathbf{1}}_{B_d}(k)\right)\left(\sum_{k'=0}^{p-1} \widehat{\mathbf{1}}_{B_d}(k')\right).$$

For each fixed bin B_d,

$$\sum_{k=0}^{p-1} \widehat{\mathbf{1}}_{B_d}(k) = \sum_{k=0}^{p-1} \sum_{r \in B_d} e^{-2\pi i k r/p} = \sum_{r \in B_d} \sum_{k=0}^{p-1} e^{-2\pi i k r/p}.$$

Since $B_d \subset \{1, \ldots, p-1\}$, every $r \in B_d$ satisfies $r \not\equiv 0 \pmod p$. The standard orthogonality of additive characters on $\mathbb{Z}/p\mathbb{Z}$ [5] gives $\sum_{k=0}^{p-1} e^{-2\pi i k r/p} = 0$ for any such r. Hence the inner sum vanishes for every $r \in B_d$, so $\sum_k \widehat{\mathbf{1}}_{B_d}(k) = 0$ for each d. The grand total is therefore a sum of products of zeros, and equals zero. □

3. The Autocorrelation Formula

Theorem 3. *For a prime p where b is a primitive root ($L = \mathrm{ord}_p(b) = p-1$), the cyclic autocorrelation satisfies*

$$R(\ell) = \frac{1}{p} \sum_{k=0}^{p-1} G\big(k,\ -(k \cdot b^{-\ell}) \bmod p\big)\,.$$

Proof. By definition, the autocorrelation counts position-wise digit matches between the repetend of $1/p$ and its rotation by ℓ, which (using $\delta(r) = \delta(s) \iff \sum_d \mathbf{1}_{B_d}(r)\mathbf{1}_{B_d}(s) = 1$) is

$$R(\ell) = \sum_{r=1}^{p-1} \sum_{d} \mathbf{1}_{B_d}(r)\, \mathbf{1}_{B_d}(b^\ell r \bmod p).$$

Apply the inverse additive Fourier expansion of each bin indicator,

$$\mathbf{1}_{B_d}(r) = \frac{1}{p} \sum_{k=0}^{p-1} \widehat{\mathbf{1}}_{B_d}(k)\, e^{2\pi i k r/p},$$

to both factors. After interchanging the order of summation and collecting the r-dependence into a single phase,

$$R(\ell) = \frac{1}{p^2} \sum_{k,k'} G(k,k') \sum_{r=1}^{p-1} e^{2\pi i (k + k' b^\ell) r/p}.$$

The inner sum is the standard nontrivial-character sum on $(\mathbb{Z}/p\mathbb{Z})^*$, namely

$$\sum_{r=1}^{p-1} e^{2\pi i m r/p} = \begin{cases} p-1 & \text{if } m \equiv 0 \pmod p, \\ -1 & \text{otherwise,} \end{cases}$$

applied with $m = k + k'b^\ell \bmod p$. The condition $k + k'b^\ell \equiv 0 \pmod p$ is equivalent (multiplying by $b^{-\ell}$) to $k' \equiv -kb^{-\ell} \pmod p$. Define

$$\Sigma_0(\ell) = \sum_{k=0}^{p-1} G\big(k,\, -kb^{-\ell} \bmod p\big)\,, \qquad \Sigma_1(\ell) = \sum_{(k,k')} G(k,k')$$

where the sum defining Σ_1 runs over the complementary pairs (k,k') with $k' \not\equiv -kb^{-\ell} \pmod p$. Substituting,

$$R(\ell) = \frac{1}{p^2}\big[(p-1)\,\Sigma_0(\ell) - \Sigma_1(\ell)\big].$$

The two sums $\Sigma_0(\ell)$ and $\Sigma_1(\ell)$ partition the full (k,k')-grid, so by Lemma 2,

$$\Sigma_0(\ell) + \Sigma_1(\ell) = \sum_{k,k'} G(k,k') = 0,$$

which gives $\Sigma_1(\ell) = -\Sigma_0(\ell)$. Substituting back,

$$R(\ell) = \frac{(p-1)\,\Sigma_0(\ell) + \Sigma_0(\ell)}{p^2} = \frac{p\,\Sigma_0(\ell)}{p^2} = \frac{\Sigma_0(\ell)}{p}. \qquad \square$$

The cancellation $\Sigma_1(\ell) = -\Sigma_0(\ell)$ is the key step. It follows from the vanishing total sum of G and eliminates the correction term entirely, leaving a clean single sum. The numerical observation that $(p-1) - (-1) = p$ exactly cancels the p^2 in the denominator is not coincidence but a consequence of the dichotomy in the standard nontrivial-character sum on $(\mathbb{Z}/p\mathbb{Z})^*$.

4. Worked Example: $p = 7$ in Base 10

We verify Theorem 3 at $p = 7$ in base 10, where $b = 10$ has order $\mathrm{ord}_7(10) = 6 = p - 1$, so the primitive-root hypothesis holds. The bins of the digit function are six singletons,

$$B_1 = \{1\},\ B_2 = \{2\},\ B_4 = \{3\},\ B_5 = \{4\},\ B_7 = \{5\},\ B_8 = \{6\},$$

and the bins B_0, B_3, B_6, B_9 are empty. Each nonempty bin contains a single residue, so its Fourier coefficient is the simple exponential

$$\widehat{\mathbf{1}}_{B_d}(k) = e^{-2\pi i k r_d/7},$$

where r_d is the residue of bin d. Summing over the six nonempty bins,

$$G(k,k') = \sum_{d\in\{1,2,4,5,7,8\}} e^{-2\pi i(k+k')r_d/7} = \sum_{r=1}^{6} e^{-2\pi i(k+k')r/7}.$$

By the standard nontrivial-character sum,

$$G(k,k') = \begin{cases} 6 & \text{if } k+k' \equiv 0 \pmod 7, \\ -1 & \text{otherwise.} \end{cases}$$

Apply Theorem 3 with $\ell = 1$. The inverse of $b = 10 \equiv 3 \pmod 7$ is $b^{-1} \equiv 5 \pmod 7$, so the orbit path is $k' = -5k \bmod 7$. Tabulating $k + k' \bmod 7$ for $k = 0, 1, \ldots, 6$:

k	$k' = -5k \bmod 7$	$k + k' \bmod 7$	$G(k,k')$
0	0	0	6
1	2	3	-1
2	4	6	-1
3	6	2	-1
4	1	5	-1
5	3	1	-1
6	5	4	-1

The sum is $\Sigma_0(1) = 6 + 6 \cdot (-1) = 0$, so $R(1) = 0/7 = 0$. Direct counting confirms this: the repetend of $1/7$ in base 10 is 142857, and at lag $\ell = 1$ it shifts to 428571, which agrees with 142857 at zero positions. The same calculation at lags $\ell = 2, 3, 4, 5$ gives $\Sigma_0(\ell) = 0$ in each case, hence $R(\ell) = 0$ for every $\ell \in \{1, 2, 3, 4, 5\}$, and direct counting confirms this as well. Together with $R(0) = 6$, this is the digit-partitioning autocorrelation of $p = 7$ predicted by Corollary 2.2 of [3].

5. The Complete Formula Chain

The autocorrelation formula closes the chain from the digit function to the eigenvalue spectrum:

(1) **Bin sizes** n_d: determined by p and b (the alignment limit paper [1]).
(2) **Bin Fourier transforms** $\widehat{\mathbf{1}}_{B_d}(k)$: Dirichlet kernels with phases from bin starting positions (the spectral power paper [4]).
(3) **Cross-spectral function** $G(k,k')$: products of bin transforms (the present paper).
(4) **Autocorrelation** $R(\ell) = \Sigma_0(\ell)/p$: G evaluated along the orbit path $k' = -kb^{-\ell} \bmod p$ (the present paper).
(5) **Eigenvalues** $\lambda_j = (1/L) \sum_\ell R(\ell)\, e^{2\pi i j \ell / L}$: DFT of R (the spectral structure paper [3]).

Every step is explicit and computable from p and b alone. The bin starting positions, which appear as the phase factors $e^{-2\pi i k a_d/p}$ in $\widehat{\mathbf{1}}_{B_d}(k)$, are determined by the bin sizes through cumulative summation

($a_d = 1 + \sum_{d'<d} n_{d'}$), so the entire chain is a function of the pair (p, b) alone. No empirical fitting, no orbit enumeration, no position-by-position digit comparison is required.

6. The Multi-Orbit Case

When $L < p - 1$, the orbit $\langle b \rangle$ is a proper subgroup of $(\mathbb{Z}/p\mathbb{Z})^*$, and $R(\ell)$ sums only over the L orbit elements. The primitive-root proof does not extend directly: the sum over the restricted orbit involves additive characters evaluated on a multiplicative subgroup (incomplete Gauss sums), which do not reduce to the two-valued dichotomy used above.

The correct generalization requires a subgroup projector involving multiplicative characters of the quotient group $(\mathbb{Z}/p\mathbb{Z})^*/\langle b \rangle$. This is not pursued here. The multi-coset autocorrelation formula remains open.

7. Remarks

G as the complete invariant. For primitive-root primes, the cross-spectral function $G(k, k')$ determines the autocorrelation $R(\ell)$ and hence the eigenvalue spectrum of the cross-alignment matrix. The function G is symmetric in its arguments, $G(k, k') = G(k', k)$, since the defining product $\widehat{\mathbf{1}}_{B_d}(k)\widehat{\mathbf{1}}_{B_d}(k')$ is commutative. It is not Hermitian in general, however: $\overline{G(k', k)} = G(-k', -k) \neq G(k, k')$ unless the bin partition is symmetric under the involution $r \mapsto -r$ on $(\mathbb{Z}/p\mathbb{Z})^*$. The spectral power $\Phi(k) = G(k, -k)$ from [4] captures the magnitude of the bin Fourier coefficients. The off-diagonal entries $G(k, k')$ for $k \neq -k$ capture the phase relationships. Together, the on- and off-diagonal data determine $R(\ell)$ via Theorem 3 and through it the eigenvalues of $\mathbf{A}(p)$.

The orbit path in (k, k') space. The map $\ell \mapsto (k, -kb^{-\ell} \bmod p)$ traces a path through the (k, k') plane for each fixed k. The autocorrelation $R(\ell)$ is the sum of G along these paths. Different primes produce different path geometries, which is why the eigenvalue spectrum varies across primes even when the bin sizes are similar.

From character sums to the program. The formula $R(\ell) = \Sigma_0(\ell)/p$ is a character sum: it evaluates the cross-spectral function at points determined by the multiplicative structure of $(\mathbb{Z}/p\mathbb{Z})^*$. This confirms the connection between the fractional field's coherence geometry and harmonic analysis suggested in earlier papers, and completes the bridge from the digit function's Euclidean structure to the additive character theory of cyclic groups [6]. The classical theory of character sums and their applications to arithmetic functions is developed in [7, 8].

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